Deep and Vision–Language Learning for Disease Classification from Chest X-rays

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Advanced Machine Learning – Assignment 1

1 Summary of Research Articles for Deep and Vision–Language Learning for Disease Classification from Chest X-rays

No.	Paper title, author(s) & year	Research problem	Proposed method	Datasets used	Experimental results	Limitations and future scope
1	XProtoNet: Diagnosis in Chest Radiography with Global and Local Explanations, Kim et al. (2021) [1]	Although deep-learning models perform well in chest X-ray diagnosis, their decisions are often difficult to understand. The study aims to make both the prediction and its supporting visual evidence interpretable.	XProtoNet learns prototype patterns for each disease and identifies where similar patterns appear in a patient's X-ray. The matched prototypes and image regions are then used to explain the final prediction.	ChestX-ray14	XProtoNet achieved mean AUC scores of 0.820 using ResNet-50 and 0.822 using DenseNet-121 on ChestX-ray14. It also reached 0.85 mean AUC without requiring bounding-box supervision.	The experiments used a single dataset, so performance across hospitals and patient groups remains uncertain. Future research should include external validation and clinical assessment of the explanations.

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2	GLoRIA: A Multimodal Global–Local Representation Learning Framework for Label-Efficient Medical Image Recognition, Huang et al. (2021) [2]	Training medical image models requires many expert-labeled images, which are costly and difficult to collect. The study aims to learn useful representations from radiology reports while preserving small, disease-related image details.	GLoRIA learns from paired chest X-rays and reports using global and local contrastive learning. It aligns complete images with reports and connects individual clinical words with the image regions they describe.	CheXpert, RSNA, SIIM-ACR	GLoRIA achieved an AUROC of 88.1% on CheXpert and 88.6% on RSNA with full training data. Using only 1% labeled data, it still achieved 86.6% and 86.1% AUROC, respectively.	The model was pretrained only on CheXpert, while its zero-shot classification relied on manually prepared clinical descriptions. Future work should evaluate it on more diverse datasets and develop automatic prompting methods.
3	IEViT: An Enhanced Vision Transformer Architecture for Chest X-ray Image Classification, Okolo et al. (2022) [3]	Standard Vision Transformers divide an X-ray into separate patches, which can reduce information about the complete image. The study investigates whether preserving this overall context can improve disease classification.	IEViT adds a CNN block that creates a representation of the complete X-ray and passes it through every transformer layer alongside the image patches. This combines local patch information with the overall image context.	Pneumonia, Tuberculosis, COVID-19 Radiography, COVIDx	IEViT achieved F1-scores of 98.48% for pneumonia, 100% for tuberculosis, 98.05% for COVID-19 and viral pneumonia, and 96.39% on COVIDx. It consistently outperformed the corresponding standard ViT models.	IEViT requires more parameters than standard ViT and was mainly tested on binary or three-class problems. Future work should reduce its computational cost and evaluate multi-label classification using pretraining on chest X-rays.

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4	Expert-Level Detection of Pathologies from Unannotated Chest X-Ray Images via Self-Supervised Learning, Tiu et al. (2022) [4]	Chest X-ray classifiers usually depend on large labeled datasets and cannot recognize diseases that were not labeled during training. The study aims to perform multi-label diagnosis without manually annotated training images.	CheXzero uses contrastive learning to align chest X-rays with their radiology reports. During zero-shot classification, it compares each image with positive and negative disease descriptions to predict whether a condition is present.	MIMIC-CXR, CheXpert, PadChest	Without labeled training examples, CheXzero achieved a mean AUC of 0.889 for five CheXpert conditions and performed comparably to radiologists in mean MCC and F1-score. On PadChest, it achieved AUC above 0.90 for 6 of 57 findings.	The method still requires labeled validation data for threshold selection and depends on the terminology used in reports. Future work should support higher-resolution images, smaller abnormalities, other imaging modalities, and non-image clinical data.
5	Multi-Granularity Cross-Modal Alignment for Generalized Medical Visual Representation Learning, Wang et al. (2022) [5]	Existing medical vision–language models mainly focus on matching images and reports, without fully considering detailed regions or disease-level similarities. The study aims to learn richer representations that transfer well to different tasks.	MGCA aligns chest X-rays and reports at three levels: complete image–report pairs, local visual and text tokens, and disease-related groups. These complementary alignments produce representations that can transfer to different medical imaging tasks.	MIMIC-CXR, CheXpert, RSNA, COVIDx, Object-CXR, SIIM-ACR	With full training data, MGCA achieved 89.7% AUROC on CheXpert, 90.8% AUROC on RSNA, and 92.3% accuracy on COVIDx. It produced the best classification results across all nine data-availability settings.	The study did not evaluate MGCA on image–text retrieval, and its three alignment levels were optimized in parallel rather than jointly. Future work could unify these levels and combine classification-focused learning with automatic report generation.

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6	A Cross-Modal Deep Metric Learning Model for Disease Diagnosis Based on Chest X-ray Images, Jin et al. (2023) [6]	Chest X-ray models usually need labeled examples of every disease, which is not practical for newly emerging or rare conditions. The study aims to identify multiple diseases, including those not available during training.	The model learns a common space for chest X-ray features and textual descriptions of diseases. It uses a weighted distance function that focuses more strongly on difficult cases and links each image with its most relevant disease labels.	ChestX-ray14	On ChestX-ray14, the model obtained a mean AUROC of 0.80 for known diseases and 0.68 for unseen diseases. Its harmonic mean reached 0.735, improving the unseen-class result of the strongest comparison method.	The experiments were limited to ChestX-ray14, so performance on other hospitals and datasets remains unknown. Future work could test few-shot diagnosis of rare diseases and combine convolutional networks with transformers.
7	MedKLIP: Medical Knowledge Enhanced Language–Image Pre-Training for X-ray Diagnosis, Wu et al. (2023) [7]	Medical vision–language models often learn directly from complex radiology reports without fully understanding clinical terms. The study aims to include medical knowledge so the model can diagnose and locate both known and previously unseen diseases.	MedKLIP converts each report into structured information about the disease, its location, and whether it is present. It expands disease terms using medical knowledge and matches this information with relevant regions of the chest X-ray.	MIMIC-CXR, ChestX-ray14, RSNA, SIIM-ACR, COVIDx, COVID Rural, Edema Severity	MedKLIP achieved zero-shot AUCs of 0.869 on RSNA, 0.892 on SIIM-ACR, and 0.768 on ChestX-ray14. For unseen COVID-19, using a detailed medical description increased the AUC from 0.656 to 0.740.	The model depends on accurate information extraction from reports and reliable descriptions from external knowledge sources. Future work could improve these components and evaluate the model using more diverse clinical datasets.

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8	Knowledge-Enhanced Visual–Language Pre-training on Chest Radiology Images, Zhang et al. (2023) [8]	Current vision–language models have difficulty identifying diseases outside their training data because they lack structured clinical knowledge. This study addresses that gap by helping the model understand relationships among medical concepts.	KAD first learns relationships between medical concepts from the UMLS knowledge graph. Its Disease Query Network then uses disease names as queries to classify findings and highlight the related regions in an X-ray.	MIMIC-CXR, PadChest, ChestX-ray14, CheXpert, ChestX-Det10	On PadChest, KAD achieved AUC above 0.90 for 31 unseen conditions and above 0.70 for 111 of 177 unseen conditions. It also outperformed the average of three radiologists on three of five CheXpert pathologies.	The model requires a labeled validation set to select prediction thresholds and provides only rough localization rather than precise lesion boundaries. Future research should improve localization and extend the approach to diseases poorly represented in reports or knowledge graphs.
9	BoMD: Bag of Multi-label Descriptors for Noisy Chest X-ray Classification, Chen et al. (2023) [9]	Large chest X-ray datasets are commonly labeled using information extracted from reports, but these labels are not always correct. Since one X-ray may contain several diseases, identifying and correcting such errors is especially challenging.	BoMD represents each X-ray using a group of visual descriptors linked to medical word embeddings. It finds potentially incorrect annotations and replaces them with smoother labels estimated from visually similar cases.	ChestX-ray14, CheXpert, OpenI, PadChest, NIH-Google, NIHx-PDC	After training on NIH ChestX-ray14, BoMD obtained mean AUCs of 89.57% on OpenI and 86.45% on PadChest. These results exceeded the closest competing methods by 1.40 and 0.96 percentage points, respectively.	The method takes longer to train because it has several stages, and label correction becomes less reliable when the noise level is extremely high. Future work should simplify training and address label noise together with class imbalance.

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10	Achieving Faster and Smarter Chest X-Ray Classification With Optimized CNNs, Louati et al. (2025) [10]	Accurate chest X-ray models are often large and computationally expensive, which limits their use on mobile or low-resource devices. Manually choosing an efficient CNN architecture also requires considerable time and experimentation.	The framework uses neural architecture search to automatically design a CNN for ChestX-ray14, followed by transfer learning to improve its performance. Filter pruning and quantization then remove unnecessary parameters and reduce the model size.	ChestX-ray14	The optimized model achieved 89.8% AUROC with 1 million parameters. After compression, its size was reduced to 0.5 million parameters, while AUROC decreased by only 0.9 percentage points to 88.9%.	The experiments were limited to ChestX-ray14 and did not include external clinical validation or model explanations. Future work could test MRI and CT data, add explainability, and explore more advanced search and compression methods.
11	Cross-Domain Invariant Feature Absorption and Domain-Specific Feature Retention for Domain Incremental Chest X-Ray Classification, Wang et al. (2025) [11]	Chest X-ray data often differ between hospitals because of variations in equipment, patient groups, and clinical practice. When a model is trained on data from a new hospital, it may forget the knowledge gained from previous hospitals.	CaD separates the learned information into two parts: knowledge that is common across hospitals and features that are unique to each hospital. It uses dedicated adapters for both parts and contrastive learning to improve the separation of different diseases.	ChestX-ray14, CheXpert, MIMIC-CXR, VinBigData, Open-I, PadChest	CaD achieved a mean AUC of 84.29% on the ChestXray14, CheXpert, and MIMIC-CXR sequence, with no recorded forgetting. On the five-dataset sequence, it reached 85.78% mean AUC, outperforming the next-best method by 5.76 percentage points.	The model needs a separate adapter for each new hospital, which increases its size as more datasets are introduced. Future work could develop a more compact approach that adapts to new or unknown hospitals without requiring separate components.

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12	Debiased Curriculum Adaptation for Safe Transfer Learning in Chest X-ray Classification, Liu et al. (2025) [12]	A model trained on one chest X-ray dataset may struggle with data from another hospital because disease frequencies and image distributions differ. Existing adaptation methods can also rely on incorrect pseudo-labels, leading to negative transfer.	DCA adapts the model gradually, beginning with more reliable examples before moving to difficult ones. It also reduces bias by balancing the learned features across disease classes and between the source and target datasets.	CheXpert, MIMIC-CXR, ChestX-ray14, Open-I	DCA was tested on 72 transfer tasks involving four datasets and six diseases, producing an average AUROC improvement of 3.196 percentage points over source-only training. For transfers from the highly imbalanced Open-i dataset, the average improvement was 8.623 points.	The experiments focused on six diseases and used public datasets rather than data collected during actual clinical deployment. Future studies could evaluate more conditions and test the method across additional hospitals and patient populations.
13	EVA-X: A Foundation Model for General Chest X-ray Analysis with Self-Supervised Learning, Yao et al. (2025) [13]	Most chest X-ray models depend on large labeled datasets and are designed for only one task, limiting their use in other clinical applications. The study aims to learn a general representation that can support different diseases and tasks with fewer annotations.	EVA-X is pretrained on more than 520,000 unlabeled chest X-rays using a Vision Transformer. It learns both disease-related meaning and spatial details by combining contrastive learning with masked-image modeling.	ChestX-ray14, CheXpert, MIMIC-CXR, COVIDx, RSNA, SIIM-ACR, Shenzhen	EVA-X-B achieved 83.5% mean AUC on ChestX-ray14, while the model reached 99.8% and 99.4% mean AUC on two COVID-19 datasets. With only 1% of the COVIDx-CXR-4 training data, it still obtained 95% accuracy.	EVA-X was trained only on public chest X-ray datasets, whose differences in disease prevalence and patient demographics may affect performance on new populations. Future work should include further clinical validation and extend the approach to other medical imaging tasks.

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14	RadZero: Similarity-Based Cross-Attention for Explainable Vision–Language Alignment in Chest X-ray with Zero-Shot Multi-Task Capability, Park et al. (2025) [14]	Existing medical vision–language models do not fully use the detailed information in radiology reports, and their attention maps provide limited explanations. The study aims to perform several chest X-ray tasks without requiring task-specific labels.	RadZero extracts short clinical findings from each report and aligns them with local image patches using similarity-based cross-attention. The resulting similarity maps support zero-shot classification, disease localization, and segmentation while showing which image regions match the text.	MIMIC-CXR, Open-I, PadChest, ChestX-ray14, CheXpert, ChestX-Det10, MS-CXR, SIIM-ACR, RSNA	RadZero achieved zero-shot AUROCs of 0.851 on Open-I, 0.841 on PadChest, and 0.879 on the rare-disease PadChest20 set. It also obtained the best average grounding score of 0.622 and the highest zero-shot Dice scores on the RSNA and SIIM datasets.	Performance decreased on some datasets, and the model depends on a pre-trained vision encoder, which may limit adaptation to new domains. Future work could improve generalization and extend RadZero to other medical images such as CT and MRI.
15	CR-MSNet: A Dual-Branch Multi-Scale Attention Network for Multi-Label Chest X-ray Classification, Wang et al. (2026) [15]	Chest X-rays may contain several diseases with lesions that vary greatly in size, making it difficult for a single network to capture both the whole chest structure and small abnormalities. Unequal numbers of examples for each disease also reduce performance on rare classes.	CR-MSNet uses a CoAtNet branch to learn global chest information and a ResNet18 branch to capture local lesion details. Multi-scale attention and cross-attention combine both features, while two-stage training reduces the effect of class imbalance.	ChestX-ray14, CheXpert	On ChestX-ray14, CR-MSNet achieved a mean AUC of 0.847 and produced the highest AUC for 13 of the 14 diseases. During external evaluation on six shared CheXpert classes, it obtained a mean AUC of 0.808.	The model was less effective for diseases with unclear boundaries, such as infiltration and pneumonia, and all images were resized to 224×224. Future work could add lesion and anatomical guidance, use higher-resolution images, and examine its application to CT or MRI.

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16	Med-ViX-Ray: Enhancing Explainable Chest X-ray Analysis with Clinical Knowledge Graphs, Cieri and Palomba (2026) [16]	Chest X-ray models can produce accurate predictions without showing the clinical reasoning behind them, making their decisions difficult for radiologists to trust. The study aims to connect each prediction with meaningful radiological signs and image regions.	Med-ViX-Ray adds a clinical knowledge graph of 14 diseases and 40 radiological signs to a Swin Transformer. The graph guides the model's attention, while a nudging module uses detected signs to refine predictions and provide clinically meaningful explanations.	MIMIC-CXR, VinDr-CXR, RSNA, MS-CXR	On MIMIC-CXR, the model improved micro-F1 from 0.496 to 0.558, recall from 0.466 to 0.734, and macro-AUROC from 0.745 to 0.787 compared with the SwinV2 baseline. It also achieved a macro-AUROC of 0.833 on VinDr-CXR without additional training.	The model depends on a fixed knowledge graph and signs extracted automatically from reports, so missing or incorrect information can affect its explanations and predictions. Future work could improve the graph and nudging process, add report generation, and evaluate the explanations with radiologists.
17	DARC: Dual Adjustment Reasoning with Counterfactuals for Trustworthy Chest X-ray Classification, Liao et al. (2026) [17]	Chest X-ray models may predict a disease from unrelated objects, such as pacemakers, or from another condition that commonly appears with it. These shortcuts can produce false predictions and make the model less reliable when the data change.	DARC uses one branch to remove the influence of non-disease objects and another to separate the visual evidence of diseases that often occur together. It is supported by CheXconf, a new dataset containing 40,213 pixel-level annotations of 11 common visual confounders.	ChestX-ray14, CheXpert, CheXconf	DARC achieved the best mean AUC on both evaluated benchmarks, reaching 0.857 on ChestX-ray14 and 0.907 on CheXpert. It also reduced false predictions caused by disease co-occurrence and remained more robust when artificial confounders were added.	CheXconf includes only 11 types of visual confounders, so other objects or artifacts seen in clinical chest X-rays may remain unaddressed. Future studies could expand the dataset and validate the model using images from different hospitals and clinical settings.

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18	CLEAR: An Auditable Foundation Model for Radiology Grounded in Clinical Concepts, Han et al. (2026) [18]	Most chest X-ray foundation models work as black boxes, making it difficult to understand why they make a prediction or why their performance decreases on new data. The study aims to make predictions traceable to clinical concepts that radiologists can inspect.	CLEAR learns from more than 870,000 image–report pairs and represents each X-ray using 368,294 clinical concepts extracted from radiology reports. Each prediction is broken down into the concepts that influenced it, allowing users to identify and correct misleading associations.	MIMIC-CXR, CheXpert-Plus, ReXGradient, VinDr-CXR, PadChest, Open-I	CLEAR achieved a mean zero-shot AUROC of 78.2% on VinDr-CXR and 70.0% on PadChest, outperforming the compared vision–language models. With supervised linear probing, it obtained an average AUROC of 87.0% across the evaluated CheXpert findings.	The concept explanations show statistical associations rather than proving causal or spatial relationships, and their quality depends on the reports and language model used to create them. Future work could train CLEAR on multilingual data from more countries and test it across a wider range of clinical settings.

2 Synthesis

Research on chest X-ray analysis has gradually moved from task-specific deep-learning models toward more general and transferable approaches. Conventional CNN-based models are effective at learning disease-related visual features, but their large computational requirements, dependence on labeled data, and limited interpretability can restrict clinical use. Optimized CNNs and multi-scale networks address some of these issues by reducing model size and combining global chest structure with local lesion information [10, 15]. Transformer-based models such as IEViT and EVA-X further improve representation learning by preserving global image context and learning from large collections of unlabeled chest X-rays [3, 13]. XProtoNet takes a complementary approach by comparing image regions with learned disease prototypes, allowing its predictions to be explained through clinically relevant visual patterns [1]. Together, these studies show that effective chest X-ray classification depends not only on predictive accuracy but also on computational efficiency, data efficiency, and interpretability.

Vision–language learning extends this development by using radiology reports as an additional source of supervision. GLoRIA aligns complete images with reports while also connecting local image regions with individual clinical words, whereas MGCA performs alignment at global, local, and disease-related levels [2, 5]. CheXzero demonstrates that image–report contrastive learning can recognize diseases without manually labeled training examples, while cross-modal metric learning uses textual disease descriptions to identify conditions that were not present during training [4, 6]. MedKLIP and KAD incorporate structured medical knowledge to improve the recognition and localization of both known and unseen diseases [7, 8]. More recent approaches, including RadZero, Med-ViX-Ray, and CLEAR, further connect predictions with local image evidence, radiological signs, and interpretable clinical concepts [14, 16, 18]. These methods indicate that combining visual information with clinical language can reduce dependence on manual annotation while supporting more flexible and explainable diagnostic systems.

Despite these advances, several challenges remain. Labels extracted automatically from reports may contain errors, and disease imbalance can reduce performance for less common conditions. BoMD addresses noisy annotations by detecting and correcting unreliable labels using visual and textual descriptors [9]. Differences among hospitals, imaging equipment, and patient populations also make it difficult for models to maintain consistent performance across datasets. CaD and DCA respond to this problem through domain-incremental learning and safer adaptation between source and target datasets [11, 12]. Another concern is shortcut learning, where predictions are influenced by medical devices, artifacts, or diseases that frequently occur together. DARC reduces these effects through confounder-aware and counterfactual reasoning [17]. Overall, the reviewed studies suggest that future chest X-ray systems should combine strong visual representations, medical knowledge, and reliable cross-domain learning. Broader external validation and clinical evaluation will also be necessary before these models can be used safely in real healthcare settings.

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